

OBSERVATION TASK 1

Course :	Full Stack Web Development
Course Code:	23CM4121
Observation Task No.:	1
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Roll No.:	A24126552108
Date:	25-08-2026

1. TITLE

College Information Portal using HTML and HTML5

2. AIM

To design and develop a structured **College Information Portal** using HTML and HTML5 components, demonstrating the use of basic HTML tags, text formatting, lists, hyperlinks, tables, images, forms, semantic elements, multimedia elements, and modern HTML5 input types.

3. OBJECTIVE

- To understand the basic structure and syntax of an HTML5 document.
 - To create a well-structured webpage using headings, paragraphs, lists, hyperlinks, tables, and images.
 - To design a student registration form using different HTML form controls.
 - To understand and implement HTML5 semantic elements such as <header>, <nav>, <section>, and <footer>.
 - To use HTML5 input types and multimedia elements such as audio, video, and iframe.
 - To develop a simple and informative **College Information Portal** using fundamental HTML and HTML5 concepts.
-

4. THEORY

HTML5 is the standard markup language used to create and structure webpages. It provides elements for text, lists, hyperlinks, tables, images, forms, and multimedia.

HTML5 also introduces semantic elements such as <header>, <nav>, <section>, <article>, and <footer> for meaningful page structure. Forms support controls such as radio buttons, checkboxes, dropdowns, and modern input types like email, date, number, color, and range.

The <audio>, <video>, and <iframe> elements enable multimedia and external content to be embedded directly into webpages. These features are combined in this task to develop a structured College Information Portal.

5. ALGORITHM / PROCEDURE

1. Create a valid HTML5 document using the <!DOCTYPE html> declaration.
 2. Add college information using headings, paragraphs, and text-formatting elements.
 3. Create ordered and unordered lists and add required hyperlinks.
 4. Create a student details table and insert the college logo and campus image with alternative text.
 5. Design a student registration form using different HTML form controls.
 6. Add semantic HTML5 elements such as <header>, <nav>, <section>, and <footer>.
 7. Add HTML5 input types such as Email, Date, Number, Color, and Range.
 8. Embed audio, video, and external content using appropriate HTML5 elements.
 9. Save the webpage and open it in a browser to verify all components and functionality.
-

6. PROGRAM

A : Index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>ANITS College Information Portal</title>

  <link rel="stylesheet" href="style.css">
</head>

<body>

  <!-- ===== HEADER ===== -->
```

```

<header>
  <h1>ANITS College Information Portal</h1>

  <p>
    <strong>Anil Neerukonda Institute of Technology and Sciences</strong>
  </p>

  <p>Visakhapatnam, Andhra Pradesh</p>
</header>

<!-- ===== NAVIGATION ===== -->
<nav>
  <a href="#about">About ANITS</a> |
  <a href="#students">Students</a> |
  <a href="#registration">Registration</a> |
  <a href="#media">Media</a>
</nav>

<!-- ===== PART A : BASIC HTML ===== -->
<main>

  <section id="about">

    <h2>Part A – Basic HTML Components</h2>

    <h3>About ANITS</h3>

    <p>
      Anil Neerukonda Institute of Technology and Sciences (ANITS)
      is an engineering institute located in Visakhapatnam,
      Andhra Pradesh.
    </p>

    <p>
      The college provides education and facilities for students
      in various engineering and technology disciplines.
    </p>

    <!-- Text Formatting -->

    <h3>Text Formatting</h3>

    <p>
      <b>Bold Text</b><br>
      <strong>Strong Text</strong><br>
      <i>Italic Text</i><br>
      <em>Emphasized Text</em><br>
      <u>Underlined Text</u><br>
      <mark>Highlighted Text</mark><br>
      <small>Small Text</small>
    </p>

```

<!-- Ordered List -->

<h3>Departments / Courses</h3>

- Computer Science and Engineering
- Electronics and Communication Engineering
- Electrical and Electronics Engineering
- Mechanical Engineering
- Civil Engineering

<!-- Unordered List -->

<h3>College Facilities</h3>

- Library
- Computer Laboratories
- Sports Facilities
- Hostel Facilities
- Transportation

<!-- Hyperlinks -->

<h3>Useful Links</h3>

<p>

Visit ANITS Official Website

</p>

<p>

Visit Google

</p>

</section>

<!-- ===== PART B : TABLES AND IMAGES ===== -->

<section id="students">

<h2>Part B – Tables and Images</h2>

<!-- Student Table -->

```
<h3>Student Details</h3>
```

```
<table border="1">
```

```
  <thead>
```

```
    <tr>
```

```
      <th>Roll No</th>
```

```
      <th>Name</th>
```

```
      <th>Branch</th>
```

```
      <th>Email</th>
```

```
    </tr>
```

```
  </thead>
```

```
  <tbody>
```

```
    <tr>
```

```
      <td>01</td>
```

```
      <td>R.Srivalli</td>
```

```
      <td>CSE</td>
```

```
      <td>srivalli@example.com</td>
```

```
    </tr>
```

```
    <tr>
```

```
      <td>02</td>
```

```
      <td>Rahul</td>
```

```
      <td>ECE</td>
```

```
      <td>rahul@example.com</td>
```

```
    </tr>
```

```
    <tr>
```

```
      <td>03</td>
```

```
      <td>Priya</td>
```

```
      <td>EEE</td>
```

```
      <td>priya@example.com</td>
```

```
    </tr>
```

```
    <tr>
```

```
      <td>04</td>
```

```
      <td>Arjun</td>
```

```
      <td>MECH</td>
```

```
      <td>arjun@example.com</td>
```

```
    </tr>
```

```
  </tbody>
```

```
</table>
```

```
<!-- Images -->
```

```
<h3>College Images</h3>
```

```
<figure>
```

```
  
```

```
  <figcaption>
```

```
    ANITS College Logo
```

```
  </figcaption>
```

```
</figure>
```

```
<figure>
```

```
  
```

```
  <figcaption>
```

```
    ANITS College Campus
```

```
  </figcaption>
```

```
</figure>
```

```
</section>
```

```
<!-- ===== PART C : REGISTRATION FORM ===== -->
```

```
<section id="registration">
```

```
  <h2>Part C – Student Registration Form</h2>
```

```
  <form>
```

```
    <!-- Name -->
```

```
    <label for="name">
```

```
      Full Name:
```

```
    </label>
```

```
    <input type="text" id="name" name="name" placeholder="Enter your name" required>
```

```
    <br><br>
```

```
    <!-- Email -->
```

```
    <label for="email">
```

```
      Email:
```

```
    </label>
```

```
    <input type="email" id="email" name="email" placeholder="Enter your email" required>
```

```
    <br><br>
```

```
    <!-- Password -->
```

```
<label for="password">
```

```
    Password:
```

```
</label>
```

```
<input type="password" id="password" name="password" placeholder="Enter password"
required>
```

```
<br><br>
```

```
<!-- Gender -->
```

```
<label>
```

```
    Gender:
```

```
</label>
```

```
<input type="radio" id="male" name="gender" value="Male">
```

```
<label for="male">
```

```
    Male
```

```
</label>
```

```
<input type="radio" id="female" name="gender" value="Female">
```

```
<label for="female">
```

```
    Female
```

```
</label>
```

```
<br><br>
```

```
<!-- Hobbies -->
```

```
<label>
```

```
    Hobbies:
```

```
</label>
```

```
<input type="checkbox" id="reading" name="hobbies" value="Reading">
```

```
<label for="reading">
```

```
    Reading
```

```
</label>
```

```
<input type="checkbox" id="photography" name="hobbies" value="Photography">
```

```
<label for="photography">
```

```
    Photography
```

```
</label>
```

```
<input type="checkbox" id="sports" name="hobbies" value="Sports">
```

```
<label for="sports">
```

```
Sports
</label>

<br><br>

<!-- Branch -->

<label for="branch">
  Branch:
</label>

<select id="branch" name="branch">

  <option value="">
    Select Branch
  </option>

  <option value="CSE">
    Computer Science and Engineering
  </option>

  <option value="ECE">
    Electronics and Communication Engineering
  </option>

  <option value="EEE">
    Electrical and Electronics Engineering
  </option>

  <option value="MECH">
    Mechanical Engineering
  </option>

</select>

<br><br>

<!-- Comments -->

<label for="comments">
  Comments:
</label>

<br>

<textarea id="comments" name="comments" rows="5" cols="40" placeholder="Enter
your comments"></textarea>

<br><br>

<!-- HTML5 Input Controls -->
```

```
<label for="age">
  Age:
</label>

<input type="number" id="age" name="age" min="1" max="100">

<br><br>

<label for="dob">
  Date of Birth:
</label>

<input type="date" id="dob" name="dob">

<br><br>

<label for="phone">
  Phone Number:
</label>

<input type="tel" id="phone" name="phone" placeholder="Enter phone number">

<br><br>

<label for="website">
  Website:
</label>

<input type="url" id="website" name="website" placeholder="https://example.com">

<br><br>

<!-- Buttons -->

<input type="submit" value="Submit">

<input type="reset" value="Reset">

</form>

</section>

<!-- ===== PART D : HTML5 FEATURES ===== -->

<section id="media">

  <h2>Part D – HTML5 Features</h2>
```

```

<!-- Audio -->

<h3>Audio</h3>

<audio controls>
  <source src="audio.mp4" type="audio/mpeg">
  Your browser does not support the audio element.
</audio>

<!-- Video -->

<h3>Video</h3>

<p>
  <a href="https://youtu.be/LGzFBhVqQ20" target="_blank">
    Watch ANITS College Video
  </a>
</p>

<!-- Iframe -->

<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>ANITS College Information Portal</title>

  <link rel="stylesheet" href="style.css">
</head>

<body>

  <!-- ===== HEADER ===== -->
  <header>
    <h1>ANITS College Information Portal</h1>

    <p>
      <strong>Anil Neerukonda Institute of Technology and Sciences</strong>
    </p>

    <p>Visakhapatnam, Andhra Pradesh</p>
  </header>

  <!-- ===== NAVIGATION ===== -->
  <nav>
    <a href="#about">About ANITS</a> |

```

```

<a href="#students">Students</a> |
<a href="#registration">Registration</a> |
<a href="#media">Media</a>
</nav>

<!-- ===== MAIN CONTENT ===== -->
<main>

<!-- ===== PART A : BASIC HTML ===== -->
<section id="about">

    <h2>Part A – Basic HTML Components</h2>

    <h3>About ANITS</h3>

    <p>
        Anil Neerukonda Institute of Technology and Sciences (ANITS)
        is an engineering institute located in Visakhapatnam,
        Andhra Pradesh.
    </p>

    <p>
        The college provides education and facilities for students
        in various engineering and technology disciplines.
    </p>

    <!-- Text Formatting -->
    <h3>Text Formatting</h3>

    <p>
        <b>Bold Text</b><br>
        <strong>Strong Text</strong><br>
        <i>Italic Text</i><br>
        <em>Emphasized Text</em><br>
        <u>Underlined Text</u><br>
        <mark>Highlighted Text</mark><br>
        <small>Small Text</small><br>
        H<sub>2</sub>O – Subscript Example<br>
        x<sup>2</sup> – Superscript Example
    </p>

    <!-- Ordered List -->
    <h3>Departments / Courses</h3>

    <ol>
        <li>Computer Science and Engineering</li>
        <li>Electronics and Communication Engineering</li>
        <li>Electrical and Electronics Engineering</li>

```

```
<li>Mechanical Engineering</li>
<li>Civil Engineering</li>
</ol>
```

```
<!-- Unordered List -->
<h3>College Facilities</h3>
```

```
<ul>
  <li>Library</li>
  <li>Computer Laboratories</li>
  <li>Sports Facilities</li>
  <li>Hostel Facilities</li>
  <li>Transportation</li>
</ul>
```

```
<!-- Hyperlinks -->
<h3>Useful Links</h3>
```

```
<p>
  <a href="https://anits.org/" target="_blank" rel="noopener">
    Visit ANITS Official Website
  </a>
</p>
```

```
<p>
  <a href="https://www.google.com/" target="_blank" rel="noopener">
    Visit Google
  </a>
</p>
```

```
</section>
```

```
<!-- ===== PART B : TABLES AND IMAGES ===== -->
<section id="students">
```

```
<h2>Part B – Tables and Images</h2>
```

```
<!-- Student Table -->
<h3>Student Details</h3>
```

```
<table border="1">
```

```
<thead>
  <tr>
    <th>Roll No</th>
    <th>Name</th>
    <th>Branch</th>
    <th>Email</th>
```

```
</tr>
</thead>

<tbody>

  <tr>
    <td>01</td>
    <td>R.Srivalli</td>
    <td>CSE</td>
    <td>srivalli@example.com</td>
  </tr>

  <tr>
    <td>02</td>
    <td>Rahul</td>
    <td>ECE</td>
    <td>rahul@example.com</td>
  </tr>

  <tr>
    <td>03</td>
    <td>Priya</td>
    <td>EEE</td>
    <td>priya@example.com</td>
  </tr>

  <tr>
    <td>04</td>
    <td>Arjun</td>
    <td>MECH</td>
    <td>arjun@example.com</td>
  </tr>

</tbody>

</table>
```

```
<!-- Images -->
<h3>College Images</h3>
```

```
<figure>

  <figcaption>
    ANITS College Logo
  </figcaption>
```

```
</figure>
```

```
<figure>
```

```
  
```

```
  <figcaption>
    ANITS College Campus
  </figcaption>
```

```
</figure>
```

```
</section>
```

```
<!-- ===== PART C : REGISTRATION FORM ===== -->
<section id="registration">
```

```
  <h2>Part C – Student Registration Form</h2>
```

```
  <form>
```

```
    <!-- Name -->
    <label for="name">
      Full Name:
    </label>
```

```
    <input type="text"
      id="name"
      name="name"
      placeholder="Enter your name"
      required>
```

```
    <br><br>
```

```
    <!-- Email -->
    <label for="email">
      Email:
    </label>
```

```
    <input type="email"
      id="email"
      name="email"
      placeholder="Enter your email"
      required>
```

```
  <br><br>
```

```
<!-- Password -->
<label for="password">
  Password:
</label>

<input type="password"
  id="password"
  name="password"
  placeholder="Enter password"
  required>
```

```
<br><br>
```

```
<!-- Gender -->
<label>
  Gender:
</label>
```

```
<input type="radio"
  id="male"
  name="gender"
  value="Male">
```

```
<label for="male">
  Male
</label>
```

```
<input type="radio"
  id="female"
  name="gender"
  value="Female">
```

```
<label for="female">
  Female
</label>
```

```
<br><br>
```

```
<!-- Hobbies -->
<label>
  Hobbies:
</label>
```

```
<input type="checkbox"
  id="reading"
  name="hobbies"
  value="Reading">
```

```
<label for="reading">
  Reading
</label>
```

```
<input type="checkbox"
  id="photography"
  name="hobbies"
  value="Photography">
```

```
<label for="photography">
  Photography
</label>
```

```
<input type="checkbox"
  id="sports"
  name="hobbies"
  value="Sports">
```

```
<label for="sports">
  Sports
</label>
```

```
<br><br>
```

```
<!-- Branch -->
<label for="branch">
  Branch:
</label>
```

```
<select id="branch" name="branch">
```

```
  <option value="">
    Select Branch
  </option>
```

```
  <option value="CSE">
    Computer Science and Engineering
  </option>
```

```
  <option value="ECE">
    Electronics and Communication Engineering
  </option>
```

```
  <option value="EEE">
    Electrical and Electronics Engineering
  </option>
```

```
  <option value="MECH">
    Mechanical Engineering
```

```
</option>
```

```
</select>
```

```
<br><br>
```

```
<!-- Comments -->
```

```
<label for="comments">
```

```
  Comments:
```

```
</label>
```

```
<br>
```

```
<textarea id="comments"
```

```
  name="comments"
```

```
  rows="5"
```

```
  cols="40"
```

```
  placeholder="Enter your comments"></textarea>
```

```
<br><br>
```

```
<!-- HTML5 Number Input -->
```

```
<label for="age">
```

```
  Age:
```

```
</label>
```

```
<input type="number"
```

```
  id="age"
```

```
  name="age"
```

```
  min="1"
```

```
  max="100">
```

```
<br><br>
```

```
<!-- HTML5 Date Input -->
```

```
<label for="dob">
```

```
  Date of Birth:
```

```
</label>
```

```
<input type="date"
```

```
  id="dob"
```

```
  name="dob">
```

```
<br><br>
```

```
<!-- HTML5 Telephone Input -->
```

```
<label for="phone">
```

Phone Number:

</label>

```
<input type="tel"
      id="phone"
      name="phone"
      placeholder="Enter phone number">
```


<!-- HTML5 URL Input -->

<label for="website">

Website:

</label>

```
<input type="url"
      id="website"
      name="website"
      placeholder="https://example.com">
```


<!-- HTML5 Color Input -->

<label for="color">

Favourite Color:

</label>

```
<input type="color"
      id="color"
      name="color">
```


<!-- HTML5 Range Input -->

<label for="rating">

Course Interest Level:

</label>

```
<input type="range"
      id="rating"
      name="rating"
      min="0"
      max="10"
      value="5">
```



```
<!-- Form Buttons -->
<input type="submit" value="Submit">

<input type="reset" value="Reset">

</form>

</section>
```

```
<!-- ===== PART D : HTML5 FEATURES ===== -->
<section id="media">
```

```
<h2>Part D – HTML5 Features</h2>
```

```
<!-- Audio -->
<h3>Audio</h3>
```

```
<audio controls>
  <source src="audio.mp4" type="audio/mpeg">
  Your browser does not support the audio element.
</audio>
```

```
<!-- Video -->
<h3>Video</h3>
```

```
<p>
  <a href="https://youtu.be/LGzFBhVqQ20"
    target="_blank"
    rel="noopener">
    Watch ANITS College Video
  </a>
</p>
```

```
<!-- Embedded YouTube Content -->
<h3>Embedded Content</h3>
```

```
<iframe
  width="560"
  height="315"
  src="https://www.youtube-nocookie.com/embed/LGzFBhVqQ20"
  title="ANITS College Video"
  frameborder="0"
  allow="accelerometer; autoplay; clipboard-write; encrypted-media; gyroscope; picture-
in-picture; web-share"
  referrerpolicy="strict-origin-when-cross-origin"
  allowfullscreen>
</iframe>
```

</section>

</main>

<!-- ===== FOOTER ===== -->

<footer>

<p>

© 2026 ANITS College Information Portal

</p>

<p>

Anil Neerukonda Institute of Technology and Sciences

</p>

<p>

Visakhapatnam, Andhra Pradesh

</p>

</footer>

</body>

</html>>

</main>

<!-- ===== FOOTER ===== -->

<footer>

<p>

© 2026 ANITS College Information Portal

</p>

<p>

Anil Neerukonda Institute of Technology and Sciences

</p>

<p>

Visakhapatnam, Andhra Pradesh

</p>

</footer>

</body>

</html>

7. CODE EXPLANATION

S.No.	HTML Component	Explanation
1	<!DOCTYPE html>	Declares that the document uses HTML5.
2	<html>	Defines the root element of the HTML document.
3	<head>	Contains metadata and page configuration.
4	<title>	Sets the title displayed on the browser tab.
5	<header>	Defines the header section containing the college information.
6	<nav>	Provides navigation links to different sections of the webpage.
7	<main>	Contains the main content of the webpage.
8	<section>	Divides the webpage into meaningful content sections.
9	<h1>, <h2>, <h3>	Define headings of different levels.
10	<p>	Displays text in paragraph form.
11	, <i>, <u>, , , <mark>	Apply different text-formatting styles.
12		Creates an ordered list of departments/courses.
13		Creates an unordered list of college facilities.
14	<a>	Creates hyperlinks to the ANITS website and Google.
15	<table>	Creates a table for displaying student details.
16	<tr>, <th>, <td>	Define table rows, headings, and data cells.
17		Displays the college logo and campus image.
18	alt	Provides alternative text for images.
19	<figure>, <figcaption>	Displays an image with its caption.
20	<form>	Creates the student registration form.
21	<input>	Creates different form controls such as text, email, password, number, and date.
22	<input type="radio">	Allows the user to select one gender option.

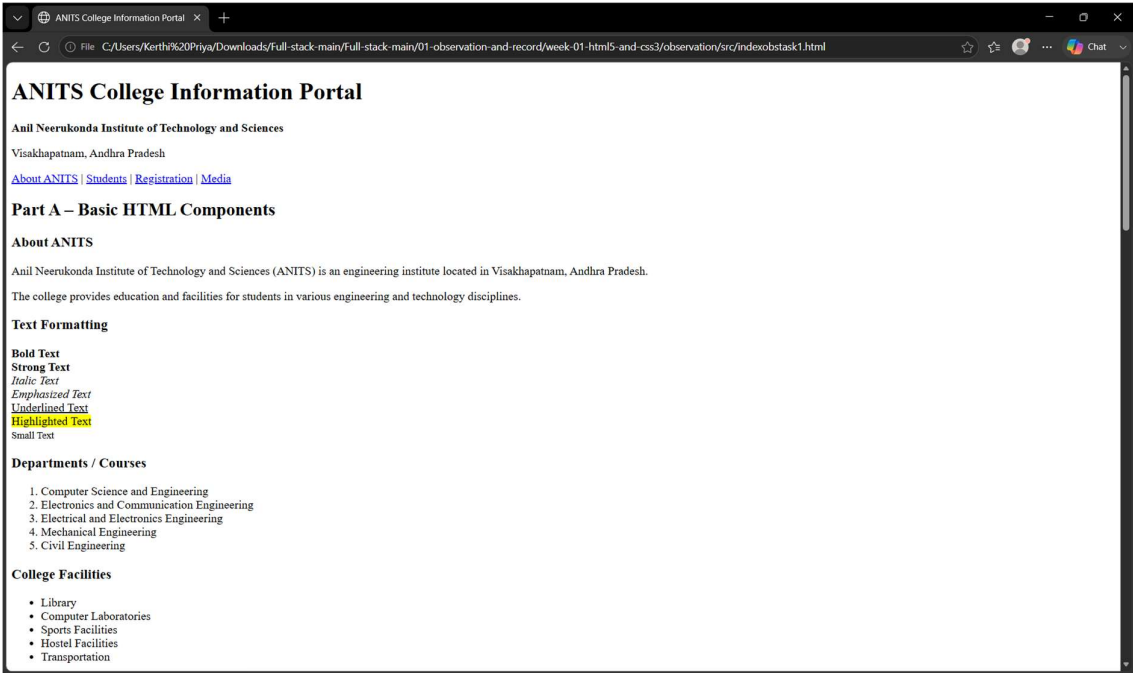
23	<input type="checkbox">	Allows the user to select one or more hobbies.
24	<select>, <option>	Creates the branch dropdown list.
25	<textarea>	Provides a multi-line field for entering comments.
26	<audio>	Embeds an audio file with playback controls.
27	<iframe>	Embeds the YouTube video within the webpage.
28	<footer>	Defines the footer containing copyright and college information.

8. TEST CASES AND EXECUTION DATA

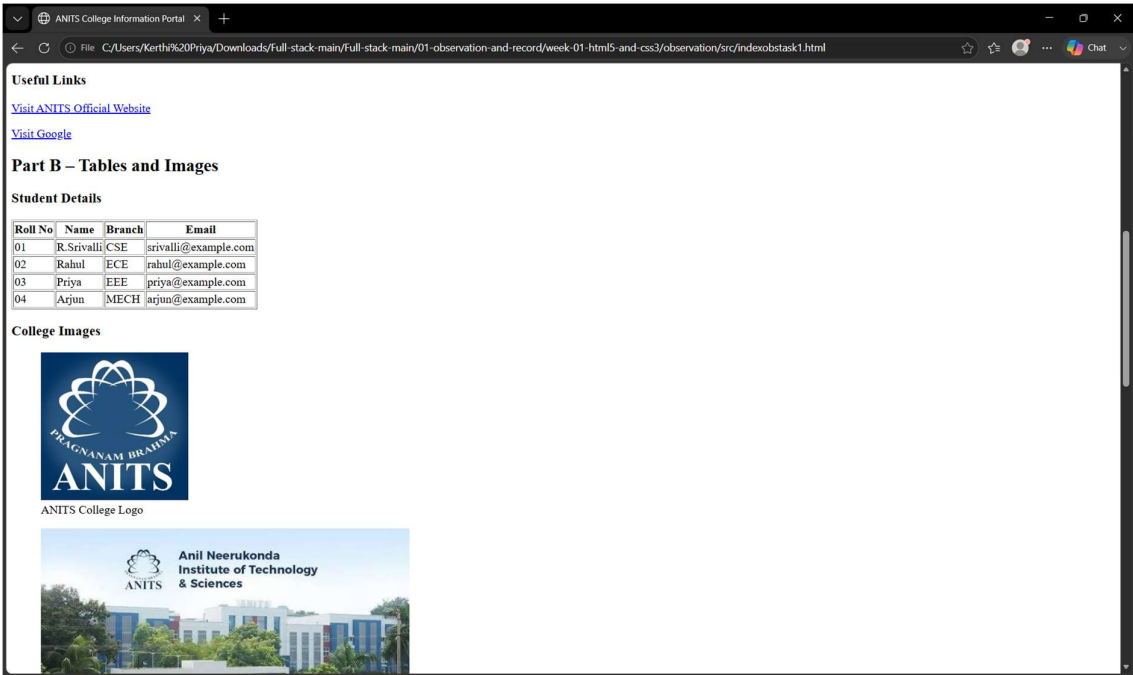
Test Case	Input / Action	Expected Result	Status
TC01	Open the HTML file in a browser	College Information Portal loads correctly	Pass
TC02	Click the navigation links	Corresponding webpage sections are displayed	Pass
TC03	Open the ANITS Official Website link	ANITS official website opens in a new tab	Pass
TC04	View the student table and images	Student details, logo, and campus image are displayed correctly	Pass
TC05	Enter valid details in the registration form	Form accepts the entered details	Pass
TC06	Leave a required field empty and submit	Browser displays a validation message	Pass
TC07	Click the Reset button	Entered form data is cleared	Pass
TC08	Play the audio / view the embedded video	Multimedia content is displayed and accessible	Pass

9. OUTPUT

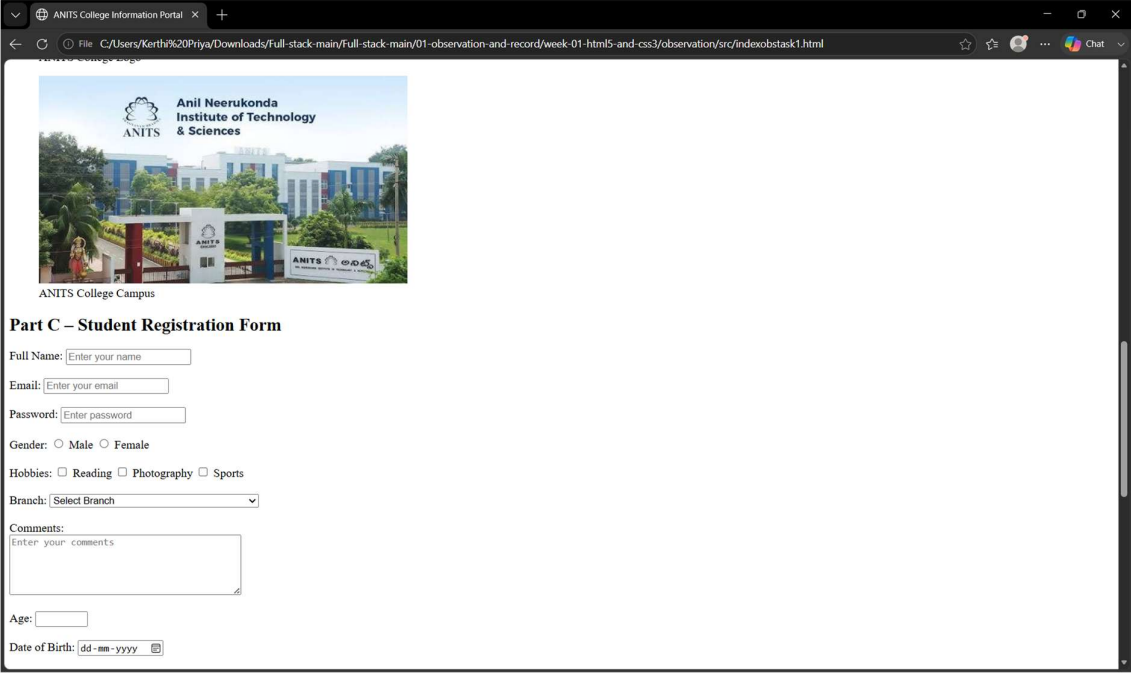
Output 1:



Output 2:



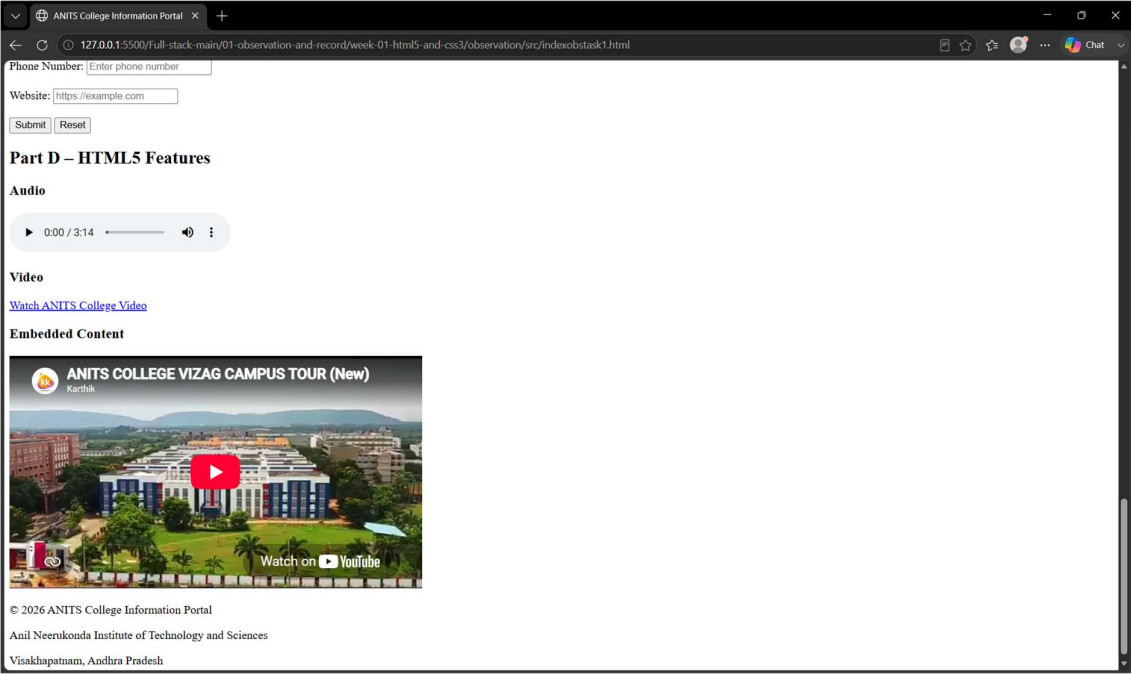
Output3:



The screenshot displays a web browser window with the title "ANITS College Information Portal". The address bar shows a local file path. The page features a header with the ANITS logo and the text "Anil Neerukonda Institute of Technology & Sciences". Below the header is a large image of the ANITS College Campus. The main content area is titled "Part C – Student Registration Form" and contains the following fields and options:

- Full Name:
- Email:
- Password:
- Gender: ☐ Male ☐ Female
- Hobbies: ☐ Reading ☐ Photography ☐ Sports
- Branch:
- Comments:
- Age:
- Date of Birth:

Output 4:



The screenshot displays a web browser window with the title "ANITS College Information Portal". The address bar shows a local file path. The page features a header with the ANITS logo and the text "Anil Neerukonda Institute of Technology & Sciences". Below the header is a large image of the ANITS College Campus. The main content area is titled "Part D – HTML5 Features" and contains the following elements:

- Phone Number:
- Website:
-
- Part D – HTML5 Features**
- Audio**
- 0:00 / 3:14
- Video**
- [Watch ANITS College Video](#)
- Embedded Content**
- 
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- Visakhapatnam, Andhra Pradesh

10. RESULT

The **College Information Portal** was successfully designed using **HTML5 components**. The webpage was created with headings, text formatting, lists, hyperlinks, tables, images, registration forms, and multimedia elements. All components were displayed and tested successfully in the web browser.